

Phase-23 Drift Surface

- Source uncertainty: `outputs/phase22/p22_uncertainty.csv`
- Drift CSV: `outputs/phase23/p23_drift_points.csv`
- Targets: [0.9, 0.94, 0.96, 0.98]
- grid: [2.7, 2.721, 2.742, 2.763, 2.783, 2.804, 2.82, 2.825, 2.846, 2.867, 2.888, 2.9, 2.908, 2.929, 2.94, 2.95] (`|grid|=16`)

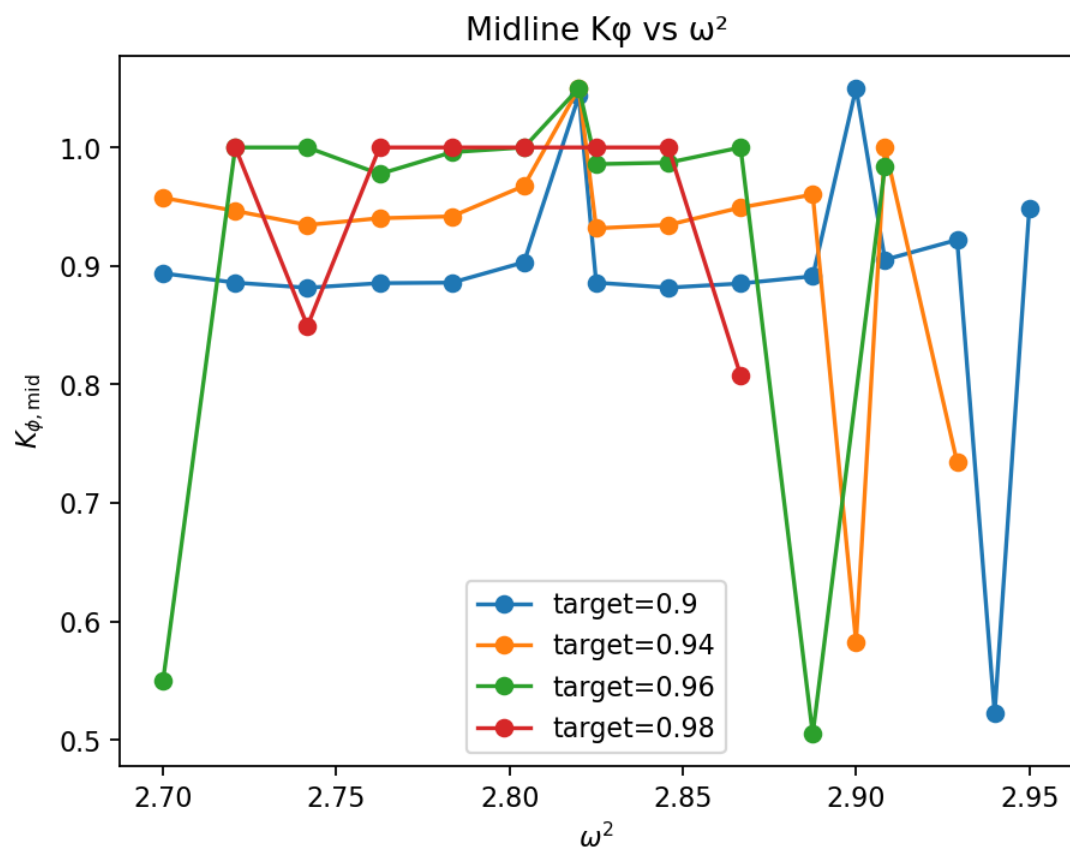
Method

For each target, we compute midline K_ϕ and estimate the drift $dK_\phi/d\omega^2$ by finite differences across ω^2 . Central differencing is used when possible; one-sided at edges.

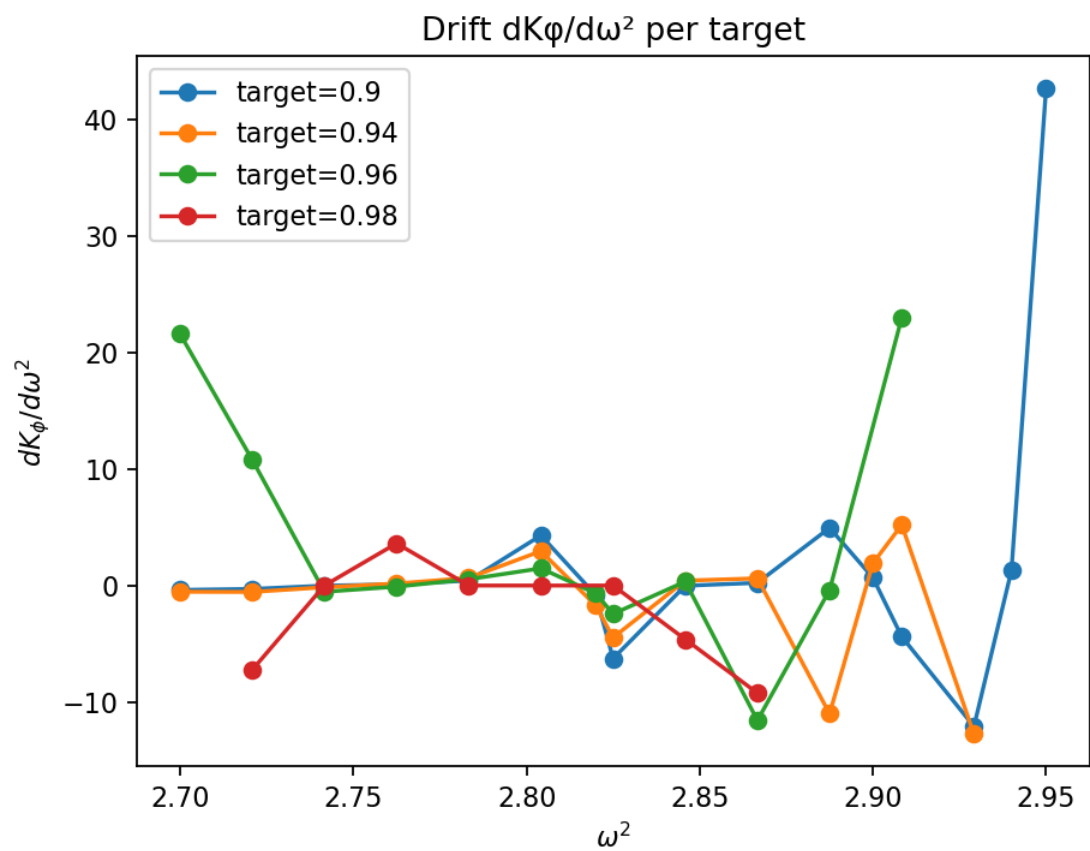
Example drift rows

target		K(mid)	dK/d
0.900	2.700	0.8938	-0.3781
0.900	2.721	0.8859	-0.2922
0.900	2.742	0.8816	-0.0107
0.900	2.763	0.8855	0.1031
0.900	2.783	0.8859	0.4232
0.900	2.804	0.9031	4.2955
0.900	2.820	1.0434	-0.8250
0.900	2.825	0.8859	-6.2596

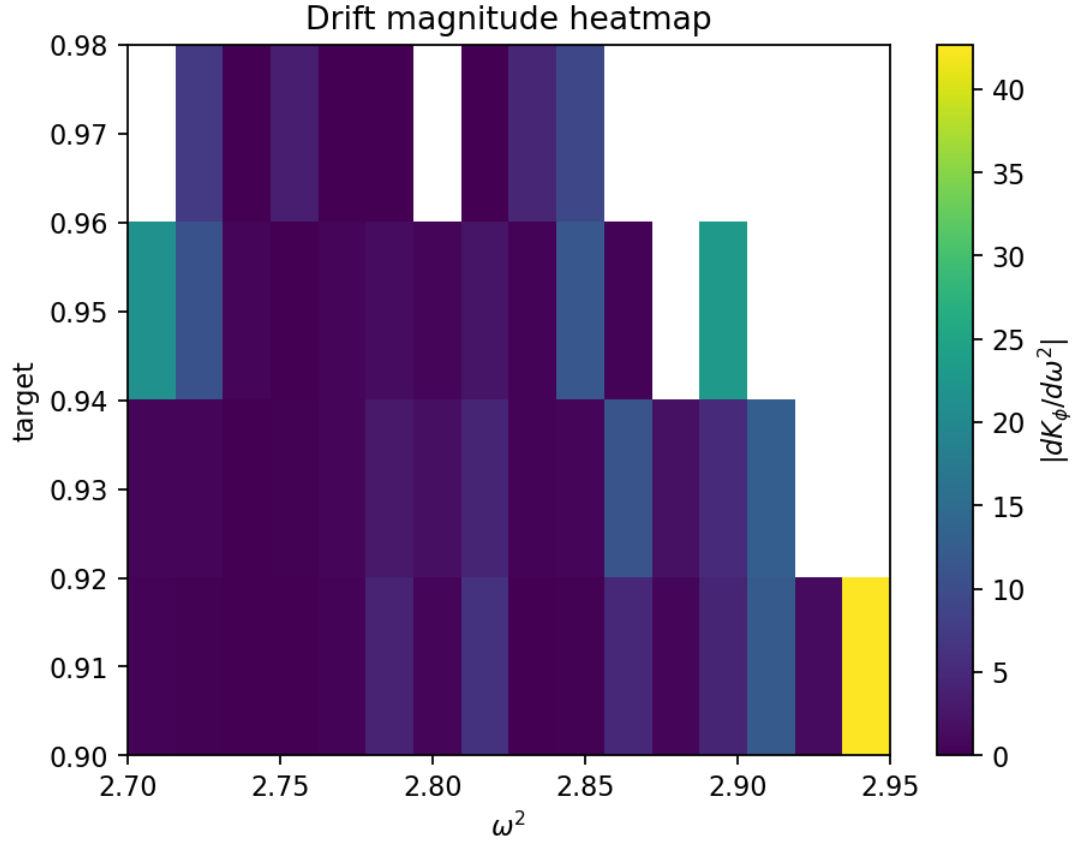
Midline K vs



Drift dK/d per target



Drift magnitude heatmap



Feasibility @ B=6

With an actuator bandwidth cap of $B = 6$ (units of $|dK_\phi/d\omega^2|$):

- 0.90 $\rightarrow$ 13/16 feasible; spikes at $\omega^2 \in \{2.83, 2.93, 2.95\}$
- 0.94 $\rightarrow$ 12/14 feasible; spikes at $\{2.89, 2.93\}$
- 0.96 $\rightarrow$ 8/12 feasible; spikes at $\{2.70, 2.72, 2.87, 2.91\}$
- 0.98 $\rightarrow$ 6/8 feasible; spikes at $\{2.72, 2.87\}$

These are narrow lock-loss precursors. Mitigations: (i) exclude those bins from the operating window, (ii) raise B to 8, or (iii) add mild hysteresis/rate-limit on K_ϕ updates.